8. Complete the frequency table below and use it to find the median value of the following set of 19 results.

11, 9, 13, 14, 11, 14, 13, 12, 13, 15, 13, 14, 11, 10, 14, 14, 15, 13, 12

	9	10	11	12	13	14	15
Frequency	1				5		

Section 7D The Mode

B: The mode
The mode of a set of numbers is the number that occurs most often.

2, 3, 3, 4, 4, 4, 5
mode

Example 3

The ages of students on a school bus are:
12 15 12 13 14 16 15 11 12 16 15 16 14 15 15
Find the mode.

Placing the ages in order we get:
11 12 12 12 13 14 14 15 15 15 15 15 16 16 16

The mode is 15 because it occurs most often (the most frequent)

Example 4

By using a tally chart/ frequency table, find the mode of this set of temperatures.
16, 18, 15, 17, 17, 18, 14, 16, 15, 16, 17, 16,

Temp (°C)	Tally	Frequency
14	|	1
15	||	2
16	||||	4
17	|||	3
18	||	2

The mode is 16°C, since it has the highest frequency

Exercise 7D

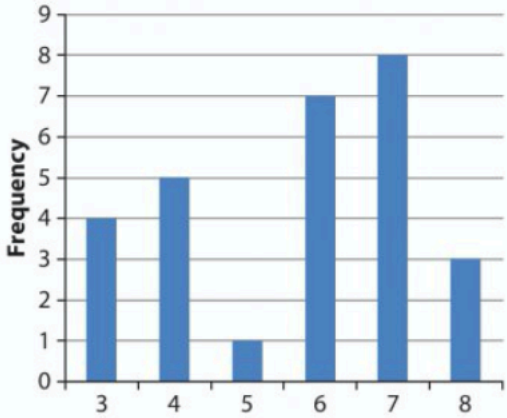
1. Find the mode of each of the following sets of data.
(i) 3, 4, 1, 2, 4, 5, 1, 3, 4, 1, 3, 1
(ii) 4, 7, 5, 6, 5, 8, 5, 7, 5,7, 5, 8, 6
(iii) 5, 6, 7, 5, 6, 7, 5, 7, 6, 6
(iv) red, blue, green, black, green, black, white, green, red, black, black
(v) A, B, A, C, B, A, C, B, B, B, C, A, C, A, A

2. After a maths test a teacher presented these results to the class.

Mark	2	3	4	5	6	7	8	9
Freq	1	2	1	4	8	6	5	3

(a) What was the modal mark?
(b) How many students were in the class?
(c) By writing out a list of all the marks or otherwise, find the median mark.

3. After a maths test the teacher presented the class results as a frequency chart as shown.
(i) What was the modal mark of the class?
(ii) How many students got the Modal mark
(iii) How many students sat the test?
(iv) Find the median class mark.



4. (i) Write out 7 numbers so that the mode is 4.
(ii) Write out 8 numbers so that the mode is 1.
(iii) Write out 7 numbers so that the median number is 3 and the mode 2.

5. Roisin did a survey on the number of absences in her year group per day over a period of 30 consecutive school days.

Her raw results (data) were:
2, 0, 4, 3, 2, 1, 5, 3, 1, 3,

4, 3, 0, 1, 5, 3, 4, 5, 2, 1,

1, 2, 1, 1, 4, 0, 1, 1, 3, 3

Number of days	Tally	Frequency
0		
1		
2		
3		
4	||||	4
5		

(i) Make a tally and frequency chart of this data.
(ii) Use the chart to find the modal number.
(iii) The median of 30 is the (15th + 16th)/2 when the data is in order. What is the median of this data?

6. A die was rolled 14 times and the number recorded.
(i) Find the mode of this data.
(ii) By listing the results in order find the median value of this data.

7. The under 10's goal record was recorded in a frequency table as shown.

(i) What was the mode of this data?
(i.e. the modal number of goals scored)

(ii) How many matches were played in total?

(iii) Find the median number of goals.

8. In Tanya's experiment with pea pods:

(i) if the modal number of peas in a pod was 11, find the smallest possible value of x.

(ii) if the median number of peas in a pod was 10, find the largest possible value of x.

No. of peas in a pod	9	10	11
No. of pods	10	12	x